INCH-POUND

MIL-C-0052980C (ME)
11 August 1994
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MIL-C-52980B
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MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING AND ROTARY, AIR DIESEL-ENGINE-DRIVEN, OR GASOLINE-ENGINE-DRIVEN

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-52980B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers diesel engine driven and gasoline engine driven, garage type, reciprocating and rotary air compressors.

1.2 Classification. The compressors shall be of the following types, classes, sizes, and styles as specified (see 6.2) . Class 2 shall be limited to the 5-cubic-feet-per-minute (cfm) size.

Type I - Reciprocating.
Type II - Rotary.
Class 1 - Tank-mounted.
Class 2 - Handtruck mounted.
Class 3 - Caster mounted.
Class 4 - Trailer mounted.
Size - 5 cfm at 175 psig.
Size - 15 cfm at 175 psig.
Size - 25 cfm at 175 psig.
Style 1 - Diesel Engine Driven (DED)
Style 2 - Gasoline Engine Driven (GED)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2) .

SPECIFICATION

FEDERAL

A-A-52155	Engine Accessory Drive, V-Belt.
FF-C-88	Casters, Rigid and Swivel, Industrial, Heavy-Duty.
GG-G-91	Gauges, Tire Pressure, Self-Contained and Inflator Gauge, Pneumatic Tire.
W-F-800	Fuel Oil, Diesel.
22-1-550	Inner Tube, Pneumatic Tire.
ZZ-T-41O	Tire, Pneumatic, Industrial.

MILITARY

MIL-V-173	Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	Treatment and Painting of Materiel.
MIL-G-3056	Gasoline, Automotive, Combat.
MIL-F-3541	Fittings, Lubrication.
MIL-C-3600	Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant) , Packaging of.
MIL-G-3859	Grease, Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-H-4607	Heater, Engine and Shelter, Ground, Type H-1.
MIL-H-7365	Hose, Air Duct, for Ground Heaters.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-B-11040	Belt, V: Engine Accessory Drive.
MIL-C-13489	Cover and End Curtains; Cargo Body (for Military Vehicles).
MIL-R-14328	Rubber Sheet; Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions) .
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic.

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MI L-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MI L-T-46791	Trailers, Cargo: 500, 2,000 and 3,000 Pound Payload, 2 Wheel, Modified Commercial.
MIL-T-83133	Turbine Fuel, Aviation Kerosene Type, Grade JP-8 .

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards For Federal Services.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STB462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	Lubrication of Military Equipment.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1226	Engine, Gasoline; Air-Cooled, 1-1/2 BHP, 4-Cycle, Military Design, Model 1A08, Installation Procedures.
MIL-STD-1300	Engine, Gasoline; Air-Cooled, 6 BHP, Military Design, Model 4A032, Installation Procedures.
MIL-STD-1366	Material Transportation Systems, Dimension and Weight Constraints, Definition of.
MIL-STD-1410	Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	Noise Limits for Army Materiel.
MIL-STD-1791	Air Transport Requirement, General Specification.
MS35644	Filler Neck Assembly, Fuel Tank-Military Vehicles.

HANDBOOK

MILITARY

MIL-HDBK-113	Guide for the Selection of Lubricants, Functional Fluids, Preservatives, Preservatives and Specialty Products for Use in Ground Equipment Systems.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

ME

13208E5819 - Kit Assembly, Starter.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: us ARMY BELVOIR RDE CTR, ATTN SATBE FMT, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC (ANSI).

B19.1 - Air Compressor Systems, Safety Standards for.

(Application for copies should be addressed to: AMERCN NATL STANDS INST, 1430 BROADWAY, NEW YORK NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.

Performance Test Code 9 - Displacement Compressors, Vacuum Pumps and Blowers.

Code IX - Welding and Brazing Qualifications.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 4814 - Automotive Spark-Ignition Engine Fuel.

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(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Steel.
D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC AGENT

National Motor Freight Classification

(Application for copies should be addressed to: AMERCN TRCKNG ASS, ATTN TRFC ORDR SECT, 2200 MILL RD, ALEXANDRIA VA 22314.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J492 - Rivets and Riveting.

(Application for copies should be addressed to: SCTY OF AUTOMTV ENGRS, INC, 400 CMNWLTH DR, WARRENDALE PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to: S MGMT CNTRL DIST, 9150 E FLAIR DR, EL MONTE CA 91731.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to: UNFRM FRGHT CLASS CMMTE, ATTN TARIFF PBLSHNG OFCR, ROOM 1106, 222 S RIVERSIDE PLZ, CHICAGO IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence, Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The air compressor shall be of the commercial classification “garage type” and shall consist of either an air-cooled piston compressor unit or air-oil cooled, rotary compressor unit with an air reservoir, air hose and power unit.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.3 Winterized first article. When winterized compressors are specified (see 3.25) in addition to non-winterized compressors, two first article models shall be furnished for examination and test. One of these first article models shall be a non-winterized compressor and one shall be a winterized compressor.

3.4 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration Prevention and control. The compressors shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the compressors may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the compressors may be newly fabricated from recovered materials to the maximum extent practicable, provided the compressors produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the compressors.

3.6 Operating temperatures. The compressors shall perform as specified herein in any ambient temperature from 120 to -25 “F for non-winterized compressors; and from 120 to -50 “F for winterized compressors (see 3.25) .

3.7 Safety and human factors engineering.

3.7.1 Safety. All rotating or reciprocating parts and other parts subject to high operation temperatures that are of such a

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nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions. The compressors shall conform to the applicable requirements of ANSI B19.1a.

3.7.2 Human factors engineering. The compressor shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472 , as applicable.

3.7.3 Noise limits. When tested in accordance with 4.5.2.11, the noise level at the operator's position and occasionally occupied positions shall not exceed Category D of MIL-STD-1474. The operator's position shall be defined to be 18 inches from the control panel. occasionally occupied areas of the set shall be defined to be anywhere within 1 meter from the perimeter of the set at locations other than the operator's position. Noise hazard caution signs shall be provided in accordance with MIL-STD-1474, 4.2, when applicable (see 6.15) , except the sign shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign. For the purpose of the noise measurement, the operator position is defined as 1.5 feet from the compressor.

3.8 Endurance and maintainability.

3.8.1 Endurance. The compressor shall produce rated capacity at rated pressure for 500 hours in operating periods of not less than 8 hours. Operating periods of not less than 4 hours during high and low temperature tests may be counted (see 4.5.2.6, 4.5.2.7, 4.5.2.18 and 4.5.2.19).

3.8.2 Maintainability. The compressor shall operate as specified herein without repair or maintenance other than the contractor's recommended normal schedule adjustments and servicing, as established by a maintenance schedule prepared and submitted by the contractor prior to test. All major assemblies and installed attachments shall be accessible for maintenance repair and replacement without removal of other major assemblies and installed attachments not normally removed. Design for maintainability shall comply with 3.7.2. Cover plates which must be removed for component adjustment shall be equipped with quick-disconnect fastenings. Cover plated shall be labeled with nomenclature for items accessible through them in accordance with MIL-STD-1472, 5.9.9.3 (see 3.7.2). All fasteners shall be of corrosion-resistance material or shall be tested to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interchangeable hardware and fastening devices. Means for drainage of lubricants, fuel, and condensate traps shall be in an accessible location and shall drain without splashing on any component or on operating personnel. The drain outlets shall be so located to

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facilitate complete drainage into a suitable container. Each maintenance, assembly or disassembly operation performed as a result of testing shall be accomplished by not more than two men using common tools and specified tools furnished with the compressor.

3.9 Air hose. The air hose shall be the commercial heavy duty type 5\16-inch inside diameter (ID), with minimum rated working pressure of not less than 200 psi. Unless otherwise specified (see 6.2), the hose shall be not less than 50 feet long. Air hose couplings shall be internal swivel fitted, having a l/4-inch, standard, straight-pipe thread and a 30-degree tapered seat and shall be fitted at one end with an inflater gauge conforming to GG-G-91, type 111, style 3.

3.10 Intake air filter and silencer The compressors shall be provided with a dry-type combination intake air filter and silencer and shall have a rainshield or a similar protective device.

3.11 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference requirements shall be in accordance with MIL-STD-461, group II of class C3.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means of lubricating.

3.12.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL (see 6.5).

3.12.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II or III. Fittings shall be located in protected positions. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, type II. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.12.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.12.4 Enclosures with integral reservoir of lubricant. Enclosures, such as gear cases and transmission housings which contain a reservoir of lubricants for lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs not less than l/2-inch pipe size, or sight gauges to determine the level of the lubricant. Each enclosure shall be equipped with a drain plug. Each enclosure shall be equipped with a means for filling the enclosure with lubricant+ The drain plug shall be located so that removal of the drain plug will result in complete drainage of the

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lubricant from the enclosure. Drain outlets shall be located to allow lubricant to drain to a suitable container when the compressor is in its normal position. Integral tubes or troughs may be used to convey drained lubricants from the drain plug to the drain outlet. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to facilitate this accessibility.

3.12.5 Delivery lubrication service. When delivered, all components, except those for which preservatives are specified in 5.1, shall be lubricated. Grease lubrication, excluding lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.13 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 composition I or II as applicable, 7 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.13.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that composition II material complies with South Coast Air Quality Management District Rule 442 - Usage of Solvents.

3.14 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on identification plates conforming to MIL-P-514, type 1, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous, protected location.

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3.14.1 Shipping data Plate. Each end item, or independent component assemblage thereof, shall have a shipping data plate attached as near as practicable to the name plate. The shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the compressor in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments. The plate shall be attached by screws, bolts, or rivets marked with the following information:

a. Length - inches.
b. Height - inches.
c. Width - inches.
d. Cube - cubic feet.
e. Gross vehicle weight - pounds.
f. Center of gravity - location, inches.

3.14.2 Stenciling The gross weight and location of the center of gravity on each end item shall be stenciled on each unit in a manner easily discernible to personnel handling the equipment.

3.15 Instruction Plates. Each compressor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the compressor. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous protected location.

3.16 Treatment and painting. The compressor, components, and parts painted in commercial practice shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2), the compressor shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383. Components which reach temperatures of 450 °F or more shall not be overcoated with MIL-C-46168 top coat.

3.16.1 Camouflage Pattern data. When specified (see 6.2 and 6.3), the contractor shall be provided camouflage pattern data.

3.17 Government-furnished Property. When specified (see 6.2 and 6.6), the following property in the quantities indicated will be furnished by the Government for the applicable size compressor (see 6.6):

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Item Number	Item Description	Identification	Compressor Size	Quantity for each compressor
1	Military design l-1/2-hp gasoline engine	Model 1A08	5 cfm	1
2	Military design 6-hp gasoline engine	Model 4A032	15 cfm	1
3	Starter and 6-hp gasoline engine	NSN 2940-00-882-3401	15 cfm	1
4	Camouflage paint pattern			As required

3.18 Government-loaned property When specified (see 6.2 and 6.7), Government-loaned property, the following property in the quantity indicated will be loaned by the Government (see 6.7).

Description	Identification	Quantity
Heater, engine and shelter, ground Type H-1	NSN 4520-00-203-0224	1 each

3.19 Type I, reciprocating.

3.19.1 Compressor. The type I compressor shall deliver the capacities of free air (see 6.8) specified in table I at a pressure of 175 pounds per square inch gauge, (psig) as determined by ASME Performance Test Code 9. The components of the compressor shall be balanced so that when the compressor is operated in conjunction with the gasoline engine, vibration shall be minimized. The compressor shall not exceed the contractor's recommended continuous operating speed. The compressor shall be air cooled. The cylinder heads or cylinder assembly (if cast integral) shall be removable. The intake and exhaust valves shall be removable. A fan shall be provided for cooling.

TABLE I. Compressor characteristics.

Size (cfm)	Minimum capacity (free air) (cfm)	Maximum operating pressure (psig)	Rated discharge pressure (psig)
5	5	200	175
15	15	200	175
25	25	200	175

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3.19.2 Intercooler and aftercooler. When a multi-stage compressor is furnished, an intercooler shall cool the air between the pressure stages and shall be located so as to be cooled by the air blast from the fan. A spring-loaded-type pressure-relief valve shall be installed between the pressure stages to protect the low-pressure cylinder in case of failure of the high-pressure cylinder valves. The temperature measured at the discharge of the air receiver shall not exceed ambient +100 °F at a pressure of 200 psig. If necessary to meet the temperature requirements specified herein, an aftercooler may be provided between the last stage of the compressor and the air receiver tank. Aftercoolers shall be air cooled.

3.19.3 Pressure tubing and fittings. All pressure tubing and fittings shall be rated to withstand 1-1/2 times their normal working pressure.

3.19.4 Drive. The drive shall consist of V-belt and grooved pulley on the power unit and compressor unit. Except for the winterized compressor (see 3.25.7) , all drive belts shall conform to A-A-52155. A means for belt removal and adjustment shall be provided.

3.20 Type II, rotary.

3.20.1 Compressor. The type II compressor shall deliver the capacities of free air specified in table 1 at a pressure of 175 psig as determined by the ASME Performance Test Code 9. The components of the compressor shall be balanced so that when the compressor is operating in conjunction with the gasoline engine, vibration shall be minimized. The compressor shall not exceed the contractor’s recommended continuous operating speed. The compressor unit shall be positive displacement and air-oil cooled. A means shall be provided to remove and recover the oil from the discharge air. When a filter-type separator is used, it shall be accessible for examination, cleaning and replacement without removal of other parts or components. To prevent hydraulic lock upon starting, a means shall be provided in the stator assembly to allow accumulated oil to pass into the sump. A pressure gauge shall be provided between the compressor unit and the separator discharge to indicate the pressure in the compressor. A nonreturn valve shall be provided either in the oil separator or in the discharge from the separator to prevent reverse flow of air from the receiver.

3.20.2 Pressure tubing fittings. All pressure tubing and fittings shall be capable of withstanding 1-1/2 times their normal working pressure.

3.20.3 Drive. The type II compressor shall be either direct drive or V-belt drive. The V-belt drive shall conform to 3.19.4. When a direct drive is used, a flexible coupling shall be installed between the power unit and compressor unit. The mounting base for the power unit shall have a means for belt removal and adjustment when V-belt drive is used.

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3.21 Class 1, tank-mounted. The compressor unit and power unit shall be mounted on a base attached to the air-receiver tank.

3.21.1 Air-receiver rank. The air-receiver tank for the class 1 compressor shall be of the horizontal type and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, section VIII. The receiver shall bear the ASME stamp and no welding or other activation shall be permitted after the receiver has been certified. Unless otherwise specified (see 6.2), the capacity of the air-receiver tanks for the 5 cfm compressors shall be not less than 60 gallons and for the 15 cfm and 25 cfm compressors shall be not less than 80 gallons. Unless otherwise specified (see 6.2), the air-receiver tank fittings shall include a globe discharge valve, a drain cock, and a O to 300-psig pressure gauge having 10 psig or smaller increments. The air-receiver tanks shall include a pressure relief valve conforming to ASME Boiler and Pressure Vessel Code. The receiver shall be provided with a steel base or not less than four mounting feet for horizontal mounting. When equipped with a steel base, openings shall be provided to permit insertion of forklift tines. Each opening shall be not less than 11 inches by 3 inches in size; openings shall be located such that the center of gravity of the compressor falls between the centers of forklift openings. A means shall be provided on the receiver for securing the air hose.

3.22 Class 2, handtruck mounted. The compressor unit and power unit shall be mounted on a lightweight, 2-wheel handtruck braced to minimize vibration. A handle shall be provided which will permit moving by one person in accordance with MIL-STD-1472, 5.9.11 (see 3.7.2), when the handle is elevated to a height of not less than 30 inches and not more than 48 inches. The class 2 compressor shall be equipped with a combination storage rack and landing legs mounted on the handle end of the truck similar to that shown on figure 1 (see 6.8). The landing legs shall be equipped with soft rubber feet so spaced as to minimize movement of the compressor during operation. The complete assemblage shall withstand vibration, falling, and bumping, encountered by this type of equipment without malfunction or damage when operated as specified herein. The dry weight of the class 2, size 5 cfm compressor shall not exceed 160 pounds.

3.22.1 Air-receiver tank. The air-receiver tank for the class 2 compressor shall have a capacity of not less than 2 gallons and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, section VIII. Air-receiver tanks with an inside diameter of 6 inches or less, which are not within jurisdiction of the ASME Boiler and Pressure Vessel Code, section VIII, shall be designed for a proof pressure of 300 psig. Unless otherwise specified (see 6.2) , the air-receiver tank fittings shall include a drain cock, and a 0- to 300-psig pressure gauge having 10 psig or smaller increments. The air receiver tanks shall include a pressure relief valve conforming to ASME Boiler and Pressure Vessel Code. The pressure gauge shall be mounted in a manner to prevent damage due to vibrations.

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3.22.2 Wheels and hubs. The wheels and hubs for class 2 compressor shall be of the drop center type to accommodate the tires specified in 3.22.3. The wheels and hubs shall be furnished with roller bearings and seals.

3.22.3 Tires and tubes The tires for the class 2 compressor shall conform to ZZ-T-410, size 4.10.6. The tires shall be 2-ply rating. Tubes shall conform to ZZ-I-550.

3.22.4 Tilt requirement. When operated at rated speed and pressure, the compressor shall deliver rated capacity when tilted 15 degrees from the horizontal in any plane.

3.23 Class 3, caster-mounted. The compressor unit and power unit shall be mounted on a steel base attached to a horizontal-type, air-receiver tank. The air-receiver tank shall be constructed as specified in 3.21.1, except the tank shall be mounted on four casters conforming to FF-C-88. Two type II (swivel) casters shall be mounted on the compressor end, and two type I (non-swivel) casters shall be mounted on the engine end.

3.23.1 Caster installation. The casters shall be mounted so that the compressor shall not overturn when pushed or pulled about on a horizontal surface or when tilted 15 degrees from the horizontal in any direction. The casters shall be attached to steel mounts on the bottom side of the receiver tank and located near each end of the tank. The distance between the centerline of the casters (measured side to side) shall be not less than 6 inches greater than the maximum diameter of the air-receiver.

3.23.2 Handle and storage rack. The class 3 compressor shall be equipped with a combination handle and storage rack mounted on the compressor end. The handle and storage rack shall be removable, and similar to that shown on figure 2 (see 6.9) . The handle shall be positioned not less than 30 inches and not more than 40 inches from the floor. The storage rack shall hold the air hose specified in 3.9. The storage rack shall be located so that the hose when stored on the rack, shall not contact any component of the compressor subject to high temperature.

3.24 Class 4, trailer mounted. The class 4 air compressor shall be mounted on a trailer as specified in MIL-T-46791, type II, size A. The commercial construction equipment (CCE) trailer chassis shall be equipped with a military compatible 12 and 24 volt system as specified. The engine compressor unit and receiver shall each be directly assembled on a common base.

3.24.1 Cover. A canvas cover for storage and protection during towing shall be provided. The materials, stitching, thread, and seam of the canvas cover shall conform to MIL-C-13489. The canvas cover shall be three dimensional to enclose the engine and compressor and shall be fastened by means of a line drawn through grommets in the seam or hem.

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3.24.2 Toolbox. A toolbox made of not less than no. 14 gauge sheet steel shall be furnished with the lid edges formed to overlap the top edges of the box to prevent the entry of dirt, show, and rain. The lid shall have a lockable, over center toggle type fastener in accordance with MIL-STD-1472, 5.9.10 (see 3.7.2). The toolbox shall be of sufficient size to hold the canvas cover, special tools, inflator gauge and publications.

3.24.3 Cranking access. Cranking shall be accomplished manually by means of an automatic rewind rope pull system. The engine shall be mounted in relation to the compressor unit so that the starter pulley is accessible with clearance for starting. The force required to manually crank the engine shall not exceed 45 pounds in accordance with MIL-STD-1472, 5.9.11.4.1 (see 3.7.2).

3.24.4 Storage rack. A storage rack shall be furnished and installed on each compressor for wrapping or looping the air hose when not in use. The rack shall be installed on the compressor so that the hose shall not come in contact with any component of the compressor subject to high temperature or abrasion.

3.24.5 Air-receiver tank. The capacity of the air-receiver tank shall be not less than 20 gallons and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, section VIII . The receiver shall bear the ASME stamp and no welding or other activation shall be permitted after the receiver has been certified. Unless otherwise specified (see 6.2), the air-receiver tank fittings shall include a spring-loaded pressure relief valve, a globe discharge valve, a drain cock, and a O- to 300-psig pressure gauge having 10 psig or smaller increments. The air-receiver tanks shall include a pressure relief valve conforming to ASME Boiler and Pressure Vessel Code.

3.24.6 Transportability. The compressor shall be transportable by highway, rail, and air and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The compressor shall be equipped with tiedown and lifting provisions.

3.24.6.1 Highway transport. When the class 4 compressor is loaded on a semitrailer/tractor, it shall meet the highway permit limits of all states.

3.24.6.2 Rail transportability. The class 4, size 15 cfm compressor shall be rail transportable in CONUS and NATO countries without -restrictions. When loaded on a 50-inch (127 cm) high rail car, the compressor shall have a dimensional profile within the AAR outline diagram and the Gabarit International de Chargement (GIC) or PPI , as specified in MIL-STD-1366. The compressor shall be capable of withstanding shock loads resulting from rail impact testing specified in 4.5.2.21.1 without failure, damage, or permanent deformation.

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3.24.6.3 Air transport. The class 4 compressor at maximum gross weight shall be roll-on/roll-off (RORO) transportable in an operational configuration by C-130 and C-141 aircraft without need of shoring. Criteria for air transports is defined in MIL-STD-1791.

3.24.6.4 Slinging provision. The class 4 (base-mounted) 15 cfm compressor shall be equipped with permanently affixed provision conforming to MIL-STD-209, class 1, type 111 or class 3, type III that shall enable the compressor-trailer and compressor to be lifted in its normal travel position. Instructions shall be stenciled above or near each provision on the base-mounted compressor warning that the lifting provision shall not be used when the compressor is mounted on the trailer chassis. The provisions shall be located so that not less than l-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Lifting provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.24.6.1. Lift points shall be labelled “LIFT HERE". The labels shall conform to the requirements of MIL-STD-1472.

3.24.6.5 Tiedown provisions. When tested as specified in 4.5.2.21.3, the compressor-trailer combination shall have permanently affixed provisions that permit fastening the compressor-trailer to floor or deck of a transport medium. Tiedown provisions shall meet the requirements of MIL-STD-209, class 2, type III or class 3, type III and with MIL-STD-209 for air transport restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.2.21.2, without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labelled "TIEDOWN HERE” and combination lift/tiedown points shall be labelled "LIFT/TIEDOWN HERE". The labels shall conform to MIL-STD-1474.

3.24.7 Mobility. The class 4, size 15 cfm compressor, chassis combination shall be capable of being towed over paved roads, unimproved surfaces and cross-country terrain at normal towing speeds.

3.25 Class 4, size 15 cfm, winterized. When specified (see 6.2), the compressor shall be winterized for operation at -50 °F. When the winterized compressor is specified, the canvas cover, as specified in 3.24.1, shall not be furnished. The winterized compressor shall be provided with the following items in addition to those specified in 3.24.

3.25.1 Enclosure. A metal enclosure shall be provided to completely enclose the top, bottom, sides, and ends of the compressor assembly, to prevent the entry of blowing snow, rain, and dust , and to maintain temperature while heating and operating the compressor at -50 °F. Provisions shall be made for opening side

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panels or doors and manually adjustable louvers at the engine end to allow ventilation for the engine and compressor unit at 120 °F without excessive overheating as specified in MIL-STD-1300 for the engine, and exceed the temperature ratings as specified by the compressor manufacture for the compressor unit. Weatherstripping shall be provided on all doors and panels. Weatherstripping shall have a protective wear coating, and both the weatherstripping and protective wear coating shall be flexible at -50 “F. The weatherstripping shall conform to MIL-R-14328. The weatherstripping shall not interfere with normal operation. Insulation shall be furnished on the-interior of the enclosure and shall be non-asbestos mastic type and not less than 3/32-inch thick.

3.25.2 Heater duct connection. A heater duct connection shall be provided on the enclosure for connection of a 12-inch-diameter duct from a heater conforming to MIL-H-4607, class 3. The duct connection and cover shall conform to MIL-H-7365, size A. The duct connection cover shall be securely fastened to the enclosure by a keeperchain or other means to prevent loss when removed. The heater duct connection shall not protrude beyond outer extremes of the enclosure more than necessary to facilitate connection of the heater duct .

3.25.2.1 Ducting. Provisions shall be made for adequate ventilation and circulation of air for the compressor, while preheating and operating under cold weather conditions. The heat from the heater shall flow through the compartment in such a manner as to warm the entire engine and compressor unit to permit starting of the compressor in not more than 30 minutes at -50 °F. Ducting or baffles shall be provided to divert heated air to the batteries and to all critical moving parts which may be restricted due to ice or congealed oil or grease. The heat shall be exhausted from the compartment through a manually operated louver in such a manner as to flow over the receiver, hose rack, and hose, warming these parts prior to starting.

3.25.3 Doors , panels , and openings. Doors or panels shall be furnished with recessed latches operable by personnel wearing heavy arctic mittens. The doors or side panels shall be of a size and location to afford accessibility to the compressor unit, control equipment, and engine for routine service, adjustment, and maintenance. Doors over 18 inches in vertical height shall be hinged horizontally in two sections. Hinges shall be the continuous piano type with brass hinge pins. The roof of the enclosure shall be removable so that installed components can be removed or replaced vertically.

3.25.4 Engine. Engines shall be in accordance with 3.26. When Government-furnished engines are specified, the engine (see 3.17 and 6.6) shall include an electric and manual starting system as specified in MIL-STD-1300. The Government-furnished starter engine (see 3.17, 6.2, and 6.6) shall be installed as shown in Drawing 13208E5819. All parts required for mounting shall be furnished by the contractor.

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3.25.5 Batteries. Storage batteries and battery box shall be furnished in accordance with MIL-STD-141O.

3.25.6 Fuel tank. A fuel tank as specified in 3.26.3 shall be provided, except that the tank shall be sufficient capacity for 8 hours of operation at rated capacity.

3.25.7 Belts. All drive belts and fan belts shall conform to MIL-B-11040.

3.25.8 Control devices. The engine shall be provided with a high cylinder-head temperature device, set to automatically stop the engine in the event of excessive cylinder-head temperatures. The control device shall be set so that the head temperature specified in MIL-STD-1300 shall not be exceeded.

3.25.9 Control Panel. A control panel shall be provided for the class 4, size 15 cfm, winterized compressor with all compressor gauges, meters, valves, switches, and other operator controls as specified herein. The control panel shall be located within the enclosure, shall be shock-mounted, and shall be accessible through a watertight, hinged cover or sufficient size to expose all controls and instruments when open. opening of the control panel cover shall not vent the engine and compressor unit compartment. Each instrument, meter and control device shall be identified by a nameplate or by legible lettering on its face. All gauges and meters shall be uniformly flush mounted. A schematic airflow diagram, operating instructions, and safety precautions shall be attached to the panel or control panel cover, so as to be visible to the operator when the control panel is in the open position. The control panel shall include the following equipment:

a. Oil pressure gauge (for engine having forced-feed lubrication above 15 psig).
b. Ammeter.
c. Fuel *gauge,* liquid level.
d. Engine hour meter.
e. Starter and ignition switches.
f. Throttle and choke, or governor controls.
g. Temperature gauge, engine oil, indicating.
h. Pressure gauge, discharge air pressure (psig).
i. Pressure gauge, compressor lubricating oil pressure (psig) (if forced feed lubricated).

3.25.10 Fuels and lubricants. When specified (see 6.2), the winterized class 4, size 15 cfm compressor shall be serviced prior to shipment with military lubricants and greases as specified in table II:

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TABLE II. Service products.

Product	Specification n	Type and Grade	NATO CODE
Gasoline, auto, combat, 91 octane	MIL-G-3056	Type II	F 46
Lubricating oil, internal combustion	MIL-L-46167	OEA	O 183
Engine, arctic Greases:			
Aircraft and instrument	MIL-G-23827		G 354
Automotive and artillery	MIL-G-10924	GAA	G 403

3.26 Diesel and gasoline engines. Unless otherwise specified (see 6.2), the engine shall be a contractor-furnished commercial engine. When the contractor furnishes a commercial engine, the engine shall be in conformance with MIL-STD-1410, except as otherwise specified herein, and shall operate on the fuels as specified in 3.26.3.1, as applicable.

3.26.1 Government-furnished engine. When a Government-furnished 1-1/2 hp engine is specified for the 5 cfm compressor, the contractor shall install the engine in accordance with MIL-STD-1226. When a Government furnished 6 hp engine is specified for the 15 cfm compressor, the contractor shall install the engine in accordance with MIL-STD-1300. When a V-belt drive is used, the engine power takeoff shall be mounted as specified in MIL-STD-1226 or MIL-STD-1300 , as applicable.

3.26.1.1 Power and speed The size 5 cfm compressor shall require not more than 1.5 hp when delivering 5 cfm, 175 psig air at an engine speed of 3600 rpm. The 15 cfm compressor shall require not more than 6 hp when delivering 15 cfm, 175 psig air at an engine speed of 3600 rpm.

3.26.2 Commercial engine. For the purpose of selecting the engine, the compressor shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, except as otherwise specified herein. The engine shall be 4-cycle and air cooled. The engine shall be furnished as a complete, self-contained, power package including hand throttle control, fail-safe governor, manual starting system and engine stop switch. If an electric starter is used, it shall have a provision for a automatic rewind manual rope starting. The engine shall be capable of operating under all condition specified herein using fuels and lubricants specified herein and in MIL-STD-1410, or the engine manufacturer’s recommended operating limits.

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3.26.2.1 Size 15 cfm and 25 cfm compressor. The engine shall be furnished with a 12-volt cranking and battery charging system in accordance with MIL-STD-1410. The battery charging alternator shall have a capacity of not less than 5 ampere for the 15 cfm compressor and not less than 15 ampere for the 25 cfm compressor. A flywheel type alternator is acceptable. A dry type air cleaner with a restriction indicator (to warn the operator when the air cleaner requires servicing) shall be furnished in accordance with MIL-STD-1410. Standard commercial batteries shall be furnished. The battery cables shall be colored black and labeled “-” for negative and colored red and labeled "+" for positive in accordance with MIL-STD-1472 (see 3.7.2). A pictorial battery connection diagram shall be provided that is readable when the batteries are being serviced.

3.26.3 Fuels and fuel tank.

3.26.3.1 Engine fuels. The diesel engine provided by the contractor or furnished by the Government shall operate on fuel conforming to both W-F-800, grade DF2 and MIL-T-83133, grade JP-8. The gasoline engine provided by the contractor or furnished by the Government shall operate on gasoline conforming to ASTM D 4814.

3.26.3.2 Fuel tank. Except as specified in 3.25.6, a fuel tank with a capacity for not less than 4 hours of operation at rated load shall be provided. Construction and mounting shall be such that vibration, tightening of the mounting devices, or contact with other components of the engine or compressor shall not cause leaks to develop. When a 10-gallon or more capacity fuel tank is furnished, a filler neck assembly conforming to MS35644 shall be provided and shall incorporate a removable strainer, and have a chain secured filler cap and a fuel tank cap gasket. When a 10-gallon capacity or less fuel tank is furnished, the filler opening shall be a minimum of 1.75 inches and shall incorporate a removable strainer and chain-secured filler cap. The filler cap and chain shall be attached to the fuel tank by means other than welding or brazing. The filler cap shall be a screw cap that shall prevent fuel from leaking when the compressor is tilted 15 degrees from the horizontal in any plane and when the handtruck-mounted compressor is elevated for mobility. The tank shall be provided with an air vent and a drain. For the diesel-engine-driven compressor, the fuel tank shall be labeled “DIESEL OR JP-8 FUEL ONLY” and for the gasoline-engine-driven compressor, the fuel tank shall be labeled "MOGAS ONLY". The labeling letter height shall be 2 inches tall and in accordance with MIL-STD-1472, 5.5.1.1. A standpipe at least l/4-inch above the bottom shall be incorporated in the fuel tank outlet to prevent the entry of dirt and foreign matter into the system. The tank and all lines shall be located or positioned to minimize fire hazard or damage resulting from spilling, overflow, or drainage. Provision shall be made for connection to an auxiliary fuel supply. The fuel tank shall be equipped with a fuel level measuring device.

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3.26.4 Engine pulley. When a V-belt drive is used, “the engine drive pulley shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members for mounting and removal of the pulley.

3.26.5 Exhaust systems The exhaust manifold, muffler, and spark arrestor shall be positioned so that exhaust gases shall be directed away from the operator’s position, fuel tank, compressor air intake, and any other component or function that may be adversely affected by exhaust gases or heat and shall remain at least 4 inches away from the fuel tank under all belt adjustments. The exhaust system shall be protected against the entry of rain and shall include drainage provisions to prevent accumulation of water and condensed vapors. When a commercial engine is used, the back pressure of the exhaust system, including the muffler and the spark arrestor, shall not exceed the maximum recommended by the manufacturer at maximum horsepower output and speed under all operating conditions specified herein. When the Government furnishes the 1-1/2 hp engine then the muffler will be furnished by the Government as part of the engine. When the Government furnishes the 6 hp engine the contractor shall provide two mufflers. For Government-furnished engines the back pressure of the exhaust system, including mufflers, shall not exceed 8 inches of water at maximum horsepower output and speed under all operating conditions specified herein.

3.26.6 Power takeoff When a V-belt drive is used for the size 15 cfm and 25 cfm compressors, the contractor shall furnish and install a power takeoff on the engine for driving the compressor unit. The power takeoff shall consist essentially of a 4-bolt flange bolted to the engine flywheel and a stub shaft or other device on which the drive pulley or clutch shall be mounted. When a direct drive is used, a flexible coupling shall be installed between the engine and compressor units. The nominal rating of the power takeoff shall be not less than 150 percent of the rated horsepower of the engine.

3.26.7 Clutch. When specified for the 15 and 25 cfm size compressors (see 6.2) , the engine shall be equipped with a clutch. The clutch shall be accessible for adjustment, replacement of parts, and lubrication (unless bearings with sealed-in lubrication are used) . Adjustment of the clutch shall not require special tools. The nominal rating of the clutch shall be not less than 150 percent of the maximum engine torque. The clutch operating lever shall be accessible to the operator and labeled as to function and shall be provided with double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (engage and disengage) in accordance with MIL-STD-1472, 5.4.3.2.1.3 (see 3.7.2).

3.26.8 Pressure regulation. The compressor shall have an adjustable, pressure actuated mechanical control of the head unloading type. The control shall be adjusted to stop compression

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of air when the tank pressure has risen to 200 psig, +0 psig, -10 psig and to start compression of air when the tank pressure has dropped to 175 ±1O psig. Means shall be provided to prevent the engine from starting against tank pressure or cylinder head pressure.

3.26.8.1 Idle device. The size 15 and 25 cfm compressor shall have a device to automatically reduce the engine speed to idle speed when the compressor is unloaded.

3.27 Interchangeability. All parts having the same part number shall be functionally and dimensional interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fits or performance.

3.28 Workmanship. All parts, components and assemblies of the compressor including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and weld parts shall be clean and free from sand, dirt, fins, pits, sprues, scales, or any damaging extraneous material. The compressor shall be free of any defect that could impair its operation and serviceability.

3.28.1 Fabrication. Steel and aluminum used in the fabrication of the compressors shall provide original quality surface finish and shall be free from kinks and sharp bends. The forming of the material shall be done by methods that shall not cause damage to the metal . Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and sharp edges and burrs shall be removed. Flame-cutting may be employed instead of shearing and sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to ensure uniformity of size and shape.

3.28.2 Welding. The surfaces of parts being welded shall free of scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation of failure when the parts connected by the welds are subjected to proof and service loading.

3.28.2.1 Welding qualification. Welders shall be qualified in accordance with ASME code IX or AWS D1.1 or D1.2 for the applicable item and material.

3.28.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Bolt holes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary.

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All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. Bolts shall not extend more than 5 threads outward from their associated nuts.

3.28.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.28.5 Machine work. Tolerances and gauges for metal fits shall conform to the limitation specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

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4.3.1 Examination. The first article compressor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article compressor shall be tested as specified in 4.5.2.1 through 4.5.2.21 as applicable for the class compressor to be furnished. Failure of any test shall be cause for rejection. A first article test log (see figure 3) shall be furnished. For diesel-engine-driven compressors, JP-8 shall be used for all first article tests.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compressor shall be examined in accordance with 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Individual. Each compressor shall be tested as specified in 4.5.2.2 and 4.5.2.22. Failure of a test shall be cause for rejection. For diesel-engine-driven compressors, JP-8 or DF-2 can be used for the quality conformance tests.

4.5 Inspection procedure.

4.5.1 Examination. The compressor shall be examined for the following defects:

101. Material not as specified (see 3.4) .
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.4.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.4.2).
104. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compressor (see 3.5) .
105. Safety devices or guards missing or not as specified (see 3.7.1).
106. Ease of maintenance not as specified as defined in ANSI B19.1 (see 3.7.1).
107. Human factors engineering not as specified (see 3.7.2).
108. Noise sign not as specified (see 3.7.3).
109. Noise level not as specified (see 3.7.3).
110. Parts, special tools, and items to be furnished with compressor missing or not as specified (see 3.8.2).
111. Air hose not as specified (see 3.9) .
112. Intake air not as specified (see 3.10).
113. Lubrication not as specified (see 3.12).
114. Lubricants not as specified (see 3.12.1).
115. Lubrication fittings not as specified (see 3.12.2).
116. Pressure-release device not as specified (see 3.12.3).
117. Enclosures with integral reservoir or lubricant not as specified (see 3.12.4).

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118. Delivery lubrication not as specified (see 3.12.5).
119. Fungus and moisture resistance not as specified (see 3.13).
120. Identification marking and instruction plates not as specified (see 3.14).
121. Treatment and painting not as specified (see 3.16).
122. Color not as specified (see 3.16).
123. Camouflage data not provided as specified (see 3.16.1).
124. Pressure tubing or fitting not rated as specified (see 3.19.3).
125. V-belts, direct drive, and pulleys not as specified (3.19.4).
126. Base-mounted compressor not as specified (see 3.21).
127. Handtruck mounted compressor not as specified (see 3.22) .
128. Air receiver tank not as specified (see 3.22.1 and 3.24.5).
129. Wheels and hubs not as specified (see 3.22.2).
130. Tires and tubes not as specified (see 3.22.3).
131. Handle and storage rack not as specified (see 3.23.2).
132. Trailer not as specified (see 3.24).
133. Cover not as specified (see 3.24.1).
134. Toolbox not as specified (see 3.24.2).
135. Cranking access not as specified (see 3.24.3).
136. Storage rack not as specified (see 3.24.4).
137. Lifting and tiedown provisions not as specified (see 3.24.6.4 and 3.24.6.5).
138. Enclosure not as specified (see 3.25.1).
139. Heater duct connection not as specified (see 3.25.2).
140. Ducting not as specified (see 3.25.2.1).
141. Doors, panels and openings not as specified (see 3.25.3).
142. Fuel tank not as specified (see 3.25.6).
143. Control panel not as specified (see 3.25.9) .
144. Fuel and lubricants not as specified (see 3.25.10).
145. Installation of Government-furnished engine not as specified (see 3.26.1).
146. Clutch not as specified (see 3.26.7).
147. Interchangeability not as specified (see 3.27).
148. Workmanship not as specified (see 3.28) .
149. Fabrication not as specified (see 3.28.1).
150. Welding not as specified (see 3.28.2).
151. Bolted connections not as specified (see 3.28.3).
152. Riveted connections not as specified (see 3.28.4).
153. Machine work not as specified (see 3.28.5) .

4.5.2 Tests.

4.5.2.1 Test conditions. Only the capacity and performance test and endurance test (see 4.5.2.4 and 4.5.2.5) shall be performed without. shelter and at the climatic conditions existing at the place of test. The compressor shall operate as specified herein without maintenance other than the contractor's recommended normal schedule maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. The compressor shall be lubricated with military oils and greases. Oils shall be those designated for use at the ambient temperature at the place of test. A first article test log similar to figure 3 shall be furnished

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tabulating all appropriate test data and observations. Safety and human factor considerations shall be evaluated throughout testing.

4.5.2.2 Air receiver tank. The air receiver tank for the compressor shall be tested in accordance with the ASME Boiler and Pressure Vessel Code, section VIII. Failure of any ASME tests shall constitute failure of this test. Presence of the ASME official code U-symbol stamped or marked on the receiver and the furnishing of a copy of the ASME data sheet (Form U-1) may be accepted as evidence that the receiver conforms to the ASME Boiler and Pressure Vessel Code. For the class 2 compressor (when furnished with an air receiver tank with an inside diameter of 6 inches or less and which is not within the jurisdiction of ASME Boiler and Pressure Vessel Code) expose the air receiver tank, exclusive of pressure gauges plus safety relief devices, to a hydrostatic pressure of 300 psig, and hold at that pressure until examined for leaks and distortion. Any evidence of leakage or permanent distortion shall constitute failure of this test.

4.5.2.3 Power consumption. Drive the compressor by a dynamometer or calibrated motor and determine the brake horsepower and speed required to deliver minimum capacity at maximum operating pressure as specified in table I. The occurrence of one of the following shall constitute failure of this test:

a. Size 5 cfm compressor, with Government-furnished engine, requiring more than 1.5 continuous brake horsepower at an engine speed of 3600 rpm.
b. Size 15 cfm compressor, with Government-furnished engine, requiring more than 6 continuous brake horsepower at an engine speed of 3600 rpm.
c. Size 5, 15 and 25 cfm compressors for commercial engines, horsepower and speed required for operation of compressor in excess of the continuous brake horsepower rating for applicable speed as recommended by the engine manufacturer.

4.5.2.4 Capacity and Performance. The compressor capacity shall be determined in accordance with the method specified in the ASME Performance Test Code 9. This test shall be performed in conjunction with and simultaneously with the endurance test (see 4.5.2.5). The air-discharge temperature shall be determined when the compressor is operating at minimum capacity at maximum discharge pressure. Any one of the following conditions shall constitute failure of this test:

a. Discharge air pressure less than 175 psig.
b. Discharge air temperature in excess of ambient +100 °F.
c. Free-air capacity less than that specified in table I.

4.5.2.5 Endurance. Operate the compressor for not less than 500 hours continually in periods of not less than 8 hours of running time. For this test, an auxiliary fuel supply system may be used to permit operation for 8 hours without refueling. If an auxiliary fuel system is not used, periods of non-operation of not more than a

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10-minute duration shall be allowed for refueling. The refueling time shall not be counted as operating time but shall not cause disallowance of the operating period during which it takes place. Oil changes and other preventive maintenance shall be performed during downtime. There shall be no replacement of parts. Adjustments provided by integral devices and repairs which require 1 man-hour or less may be made. The 500 hours of operation shall be accomplished at rated capacity and pressure. Record appropriate remarks, including a record of all stops and adjustments. Record the following data at least once during each 24-hour period. Readings shall be taken only after at least 1 hour of continuous operation:

a. Air temperature, ambient.
b. Air temperature at receiver discharge.
c. Air receiver pressure.
d. Barometric pressure.
e. Air temperature at compressor intake.
f. Air temperature upstream of nozzle.
g. Air pressure of compressor intake.
h. Nozzle throat diameter.
i. Nozzle coefficient of discharge.
j. Differential pressure across the nozzle.
k. Engine speed (rpm).
l. Compressor unit speed (rpm).
m. Compressor capacity (cfm) computed.
n. Engine intake air temperature.
o. Engine oil pressure (if pressure lubricated).
p. Engine oil sump temperature.

Completion of less than 500 hours of operation at rated capacity and pressure; discharge air temperature in excess of ambient +100 °F; requirement for replacement of parts; requirement for repairs requiring more than 1 man-hour to complete; or necessity for adjustment not provided for by integral devices; shall constitute failure of this test.

4.5.2.6 High temperature. Unless otherwise specified (see 4.5.2.18), subject the compressor to an ambient temperature of 120 ±5 °F until all components have stabilized at this temperature. ‘Then start and operate the compressor at rated capacity and pressure for not less than 4 hours while maintaining 120 ±5 °F ambient temperature. Adjustments to the engine or compressor unit shall not be permitted after the first hour of’ operation. The operating time for this test may be counted toward the accomplishment of the endurance test. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to start and operate at rated capacity, pressure and speed, discharge air temperature in excess of ambient +100 °F or requirement for replacement of parts shall constitute failure of this test.

4.5.2.7 Low temperature. Unless otherwise specified (see 4.5.2.19), subject the compressor to an ambient temperature of -25 ±5 °F until all components have stabilized at this temperature.

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Then start and operate the compressor at rated capacity, pressure and speed for not less than 4 hours while maintaining -25 ±5 °F ambient temperature. Adjustments to the engine or compressor unit shall not be permitted after the first hour of operation. The operating time for this test may be counted toward the accomplishment of the endurance test. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to operate at rated capacity, pressure and speed, or requirement for replacement of parts shall constitute failure of this test.

4.5.2.8 Cycling. Start the compressor and allow the receiver pressure to build up until the pressure-regulation control operates. Then reduce the receiver pressure until the pressure-regulation control activates to increase the pressure. Vary the pressure as above until a total of 20 high-pressure actuations and 20 low-pressure actuations have been reached. Inability of the pressure-regulation control to control the pressure as specified shall constitute failure of this test.

4.5.2.9 Pressure regulation. size 5 cfm. When an adjustable pressure-regulating valve of the pressure-relief type actuated by receiver pressure is used, at the beginning and after completion of the endurance test, start the compressor and allow the receiver pressure to build up to determine pressure required to discharge full capacity through the pressure-regulating valve. Then reduce the receiver pressure to determine the pressure at which the pressure-regulating valve reseats. Nonconformance to 3.26.8 shall constitute failure of this test.

4.5.2.10 Electromagmetic interference. The compressor shall be tested in accordance with MIL-STD-462 to determine conformance to 3.11. The contractor shall furnish the contracting officer the first article test report for evaluation. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.11 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474 , 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Nonconformance to 3.7.3 shall constitute failure of this test.

4.5.2.12 Class 2.

4.5.2.12.1 Stability. During the performance of the endurance test, measure the movement during three 8-hour periods of operation. The first period shall be between O to 25 hours, the second period between 250 and 300 hours, and the third, between 450 and 500 hours. At the beginning of each of these 8-hour periods, place the landing feet of the handtruck within circles marked, scribed, etched, or painted on the floor. The circles shall be 6 inches in diameter or

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2 inches larger in diameter than the rubber feet, whichever is larger. The compressor shall be held to the floor by gravity alone, and movement shall in no way be restricted by mechanical attachments, braces, wheel chocks, or other means integral to the compressor. Location of either or both of the rubber feet, wholly or in part, outside of or upon the circles at any time during the 8-hour period of operation shall constitute failure of this test.

4.5.2.12.2 Mobility. After completion of the 500-hour endurance test, conduct a test for mobility as follows:

a. Elevate the handle to a height of not less than 30 inches nor more than 48 inches and move the compressor not less than 100 yards turning in complete circles to the right and left during the movement. At least 50 yards of the movement shall be in a straight line. Cart the compressor over a 12-inch drop onto a concrete surface not less than three times.
b. Tilt the compressor to an angle of not less than 45 degrees by elevating one side of the frame and allowing the wheel and landing leg on the opposite side of the frame to rest on a concrete floor. Allow the elevated side to fall to a position of rest on the concrete floor without any restraining forces being applied. Do this not less than two times for each side.
c. Inability to be moved by one man, inability to be carted in a straight path, inability to be turned either to the right or the left, or fracture or permanent deformation of any component or part or leakage of gasoline from the tank shall constitute failure of this test.

4.5.2.12.3 Tilted operation. Tilt the compressor 15 degrees from the horizontal about the longitudinal axis and operate in this position for 30 minutes. This test shall be performed in each of the four directions (compressor tilted with respect to longitudinal and transverse axes) . Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system, then start and operate the compressor for 30 minutes. Any malfunction due to tilted operation, permanent deformation of any part of component, or leakage of gasoline from the tank shall constitute failure of this test.

4.5.2.13 class 3.

4.5.2.13.1 Casters. Push and pull the compressor horizontally in all directions, turn the compressor 360 degrees, and tilt the compressor in 15 degrees from the horizontal in all directions. Overturning or any condition that interferes with the maneuverability of the compressor or physical damage to the compressor shall constitute failure of this test.

4.5.2.14 Exhaust and fuel system.

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4.5.2.14.1 Engine exhaust. Measure the engine exhaust pressure from a pressure tap approximately 2 inches from the exhaust manifold flange with the compressor operating at the maximum speed and the horsepower output specified herein. Exhaust pressure in excess of that specified in 3.26.5 shall constitute failure of this test. When the engine exhaust muffler is furnished by the Government as part of the military design engine, this test is not required.

4.5.2.14.2 Fuel tank capacity. Fill the fuel tank with fuel as applicable. Start the compressor and operate at rated capacity and pressure until the engine ceases to operate due to lack of fuel. The elapsed time of operation shall be recorded. No refueling or auxiliary fuel supply shall be allowed. Elapsed operation time of less than as specified in 3.26.3.2 shall constitute failure of this test.

4.5.2.15 Road test. Tow the class 4, size 15 cfm compressor, for a distance of 100 miles at maximum safe speeds. Twenty-five miles of this travel shall be on cross-country terrain (see 6.9) and 75 miles shall be on primary roads. Operation on smooth, paved surfaces shall include not less than 15 abrupt stops from a speed of 25 mph in the forward direction, not less than five abrupt stops from a speed of 2 mph in the reverse direction, and not less than 10 sharp turns to the left and 10 to the right at not less than 25 mph. Radii of the sharp turns shall be not greater than 150 feet. Displacement or damage to welded, bolted, or riveted connections or failure to operate as specified shall constitute failure of this test.

4.5.2.16 Center of gravity test. When mounted on the trailer as specified in 3.24, determine the horizontal center of gravity of the class 4, size 15 cfm compressors. Nonconformance to 3.24 shall constitute failure of this test.

4.5.2.17 Tilted operation - class 4. Tilt the class 4, size 15 cfm compressor 15 degrees from the horizontal about the longitudinal axis and operate in that position for 30 minutes. Repeat this test in each of the four directions (compressor tilted with respect to the longitudinal and transverse axes) . Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system and then start and operate the compressor for 30 minutes in that position. Any malfunction due to tilted operation such as lack of lubrication, rough or erratic operation, or permanent deformation of any part or component shall constitute failure of this test.

4.5.2.18 High temperature. Conduct the high temperature test on the class 4, size 15 cfm compressor as specified in 4.5.2.6. The test shall be performed with the canvas cover removed on the non-winterized compressor and with all side panels, louvers, and the duct in the open position or removed on the winterized compressor. In addition to the data specified in 4.5.2.5, record the compressor unit and engine lubricating oil sump temperatures, at 30-minute intervals. Failure to start and operate at rated capacity,

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pressure, and speed, oil temperature higher than specified in MIL-STD-1300 for the engine and as specified by the contractor for the compressor unit, discharged air temperature in excess of ambient +100 °F, or requirement for replacement of parts shall constitute failure of this test.

4.5.2.19 Low temperature. Subject the class 4, size 15 cfm winterized compressor to an ambient temperature of -50 °F until the batteries and the engine and compressor unit lubricating oil temperatures have stabilized. A 30-minute preheat with a heater conforming to MIL-H-4607, class 3, or equivalent, shall be utilized prior to attempting to start at -50 °F. Start the compressor with the batteries that were cold soaked with the compressor. Outside power for cranking shall not be used. After starting, disconnect the heater, replace the duct cover, and operate the compressor at rated speed, capacity and pressure for not less than 4 hours. Record the data as specified in 4.5.2.5 at 30-minute intervals. Failure to start after the 30-minute preheat period, failure to operate at rated speed, pressure and capacity after starting, or requirement for replacement of parts, shall constitute failure of this test.

4.5.2.20 High temperature shutoff. On the winterized class 4, size 15 cfm compressor, activate the high cylinder-head temperature control device to stop the engine to determine conformance to 3.25.8. The device may be activated after complete or partial disassembly from the engine and by the application of an outside high temperature source to simulate high cylinder-head temperature.

4.5.2.21 Transportability.

4.5.2.21.1 Rail impact test. Equipment shall be tested in accordance with MIL-STD-810, method 514.4, procedure III, and method 516.4, procedure VIII, except the item shall be impacted in each direction at speeds of 6 and 8 miles per hour, ±5 percent. Examine the equipment at the completion of each phase of this test. Any dislodgement of any tool or item or equipment from its designated position, any loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the equipment shall constitute failure of this test. Movement of a tool or item of equipment within its designated position without any dislodgement is permitted.

4.5.2.21.2 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.24.6.4. Nonconformance to 3.24.6.4 shall constitute failure of this test.

4.5.2.21.3 Tiedown Provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.24.6.5. Nonconformance to 3.24.6.5 shall constitute failure of this test.

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4.5.2.21.4 Air transportability. The air transportability requirement for the C-130 and C-141 aircraft in accordance with MIL-STD-1791 (see 3.24.6.3) shall be certified. Nonconformance to 3.24.6.3 shall constitute failure of this test.

4.5.2.22 Capacity. Test each compressor for capacity in accordance with 4.5.2.4 except that each compressor shall be tested as a unit driven by its gasoline engine. Prior to testing, run-in each compressor as prescribed by the contractor. Occurrence of one or more of the following conditions shall constitute failure of this test:

a. Discharge air pressure less than 175 psig.
b. Discharge air temperature in excess of ambient temperature +100 °F.
c. Free-air capacity less than that specified in table I.

4.6 Inspection of Packaging. The presentation, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of MIL-C-3600. The presentation, packing and marking shall be examined to determine compliance with the requirements specified in section 5.

5. PACKAGING

5.1 Preservation. Preservation of each compressor accessories, repair parts, tools, and publications, shall be in accordance with level A, B, or C requirements of MIL-C-3600, as specified (see 6.2) .

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A and B. Each compressor, preserved as specified in 5.1, shall be packed in accordance with the applicable level A or B requirements of MIL-C-3600.

5.2.2 Level C. Each complete compressor and all items issued with the compressor, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules; and as a minimum compressors shipped loose or mounted on skids in lieu of being boxed or crated shall be protected with a minimum 6 mil waterproof plastic shroud. The shroud shall be placed over the compressor and secured to the compressor base. Openings in skids shall be a minimum of 3 inches in height to provide access for forklift tines. Support legs of trailer mounted compressor shall be utilized for packing and storage only and shall not be used for support of the trailer during shipment.

5.3 Marking. In addition to any special or identification markings required by the contract or purchase order (see 6.2) , marking shall be in accordance with MIL-STD-129.

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6, NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The compressors are intended for use as a source of compressed air in the normal operations conducted at filling stations and vehicle maintenance shops.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2) .
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of first article model and number of compressors to be furnished (see 3.2) .
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2 and 3.3).
f. When winterized first article examination and testing is required and number of compressors to be furnished (see 3.3).
9" Length of air hose to be furnished if other than as specified (see 3.9).
h. When electromagnetic interference requirements other than as specified in 3.11 are required.
i. When color required is other than camouflage green 383 (see 3.16).
j. When overcoat is required in accordance with camouflage pattern data (see 3.16.1).
k. When camouflage data will be provided (see 3.16.1 and 6.3).
l. When Government engines are to be furnished in place of commercial engines (see 3.17, 3.25.4, and 3.26).
m. When Government-furnished property is to be provided (see 3.17).
n. When Government-loaned property is to be provided (see 3.18).
o. Capacity of air receiver tank if other than as specified (see 3.21.1).
p. Air-receiver-tank fittings required if other than as specified (see 3.21.1, 3.22.1, and 3.24.5).
q. When winterized class 4, size 15 cfm compressors are required (see 3.25).
r. When winterized class 4, size 15 cfm compressors will be serviced prior to shipment (see 3.25.10) ,
s. When sizes 15 cfm and 25 cfm compressors will be furnished with a clutch (see 3.26.7) .
t. Whether the engines to be used shall be diesel engines or gasoline engines (see 3.26).
u. Level of preservation and packing required (see 5.1 and 5.2).

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v. Any special marking (see 5.3).
w. When a transportability report is required (see 6.3) .

6.3 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

	Paragraph No.	Data requirements	Applicable DID No.
a.	3.16.1	Color photograph prints	DI-MISC-80192
b.	3.16.1	Camouflage line art data	DI-MISC-80176
c.	3.24.6	Transportability Report	DI-PACK-80880
d.	3.28	Welding Procedure Qualification Test Report	DI-MISC-80876
e.	3.28	Welding Procedures	DI-MISC-80875
f.	4.3	First article inspection procedure	DI-T-4901
g.	4.3	First article inspection report	DI-T-4902

(Copies of DIDs required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders-offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provide specific

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requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the: us ARMY BELVOIR RD&E CTR, ATTN SATBE FL, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.6 Government-furnished Property. The contracting officer should arrange to furnish the property listed in 3.17, 3.25.4 and 3.26.5 as required. The military design engine and information regarding the installation of the engine will be furnished by the Government (see 3.26.1).

6.7 Government-loaned property The contracting officer should arrange for loan of the Government property specified in 3.18 as required.

6.8 Free air. Free air is defined as air at atmospheric conditions surrounding the compressor, at the point where the compressor is installed, but not in the immediate vicinity where it may be affected by the heat of the compressor or engine cooling system.

6.9 Information figures. Figures 1 and 2 show a handtruck-mounted compressor and a caster-mounted compressor, respectively, which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other designs which conform to this specification.

6.10 Cross-country terrain. Cross-country terrain shall consist of generally level or improved open fields, broken ground, loose sand, or gravel road (see 4.5.2.15).

6.11 Publications. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor’s maintenance and operation manual to be furnished with each compressor.

6.12 Provisioning. The contracting officer should include provisioning requirements for repair parts and special tools as necessary, and instructions on shipment of compressors. A suggested paragraph is as follows:

> “Shipment of compressors shall include repair parts, special tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract.”

6.13 Classification of materials. The contracting officer shall require the contractor to provide certification of the equipment meeting ANSI B19.1a and the ASME standards.

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6.13 Engines. Reference to gasoline-engine-driven compressors in this specification is limited to other than U.S. Army applications in accordance with DA guidance.

6.14 Subject term (key word) listing.

Air compressor, garage type
Air compressor, gasoline-engine-driven, air cooled
Air reservoir
Compressor, reciprocating, 5 cfm at 175 psig
Compressor, reciprocating, 15 cfm at 175 psig
Compressor, reciprocating, 25 cfm at 175 psig
Compressor, reciprocating, caster mounted
Compressor, reciprocating, hand-truck-mounted
Compressor, reciprocating, tank mounted
Compressor, reciprocating, trailer mounted
Compressor, rotary, 5 cfm at 175 psig
Compressor, rotary, 15 cfm at 175 psig
Compressor, rotary, 25 cfm at 175 psig
Compressor, rotary, caster mounted
Compressor, rotary, tank mounted
Compressor, rotary, trailer mounted

6.15 Noise limits. Where the limit of Category D can be documented as being clearly beyond the state-of-the-art, per MIL-STD-1474, 5.1.1.2, selection of another noise limit should be considered by the procuring activity, as described in MIL-STD-1474, acceptance requirements (5.1.1.3) . Particular attention should be given to 5.1.1.2.2.d. When permission is granted to exceed the limits of category D, the procuring activity should ensure that operator and maintenance manuals are prepared in accordance with MIL-STD-1474, manuals requirement (4.3) .

6.16 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

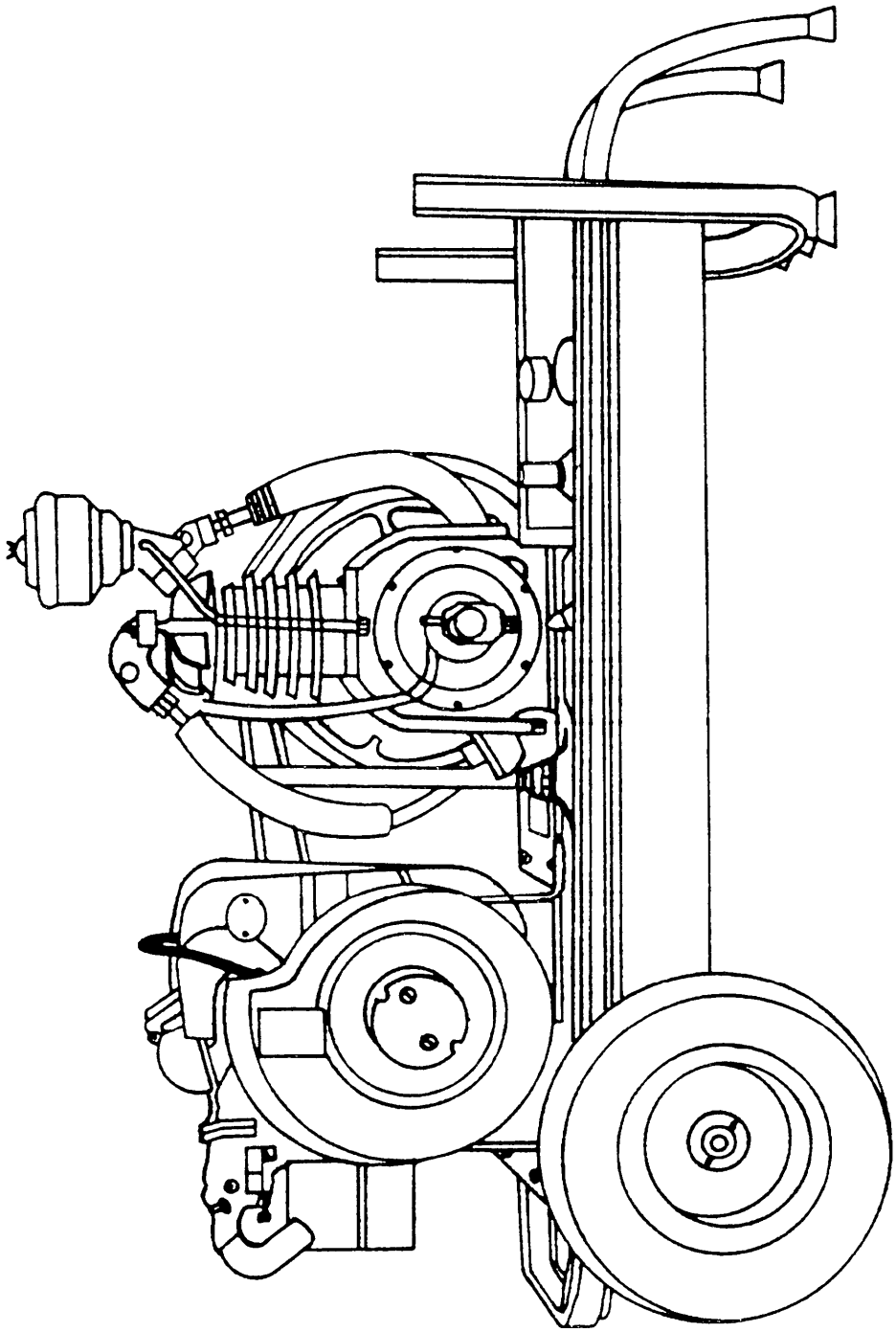
6.17 Welders and welding operators. The contracting officer should consider re-qualifying welders or welding operators in the event of evidence of poor welds (see 3.28.2.1).

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Preparing activity:
Army - ME

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X3640

FIGURE 1. Compressor, air, hand-truck-mounted (gen. configuration).

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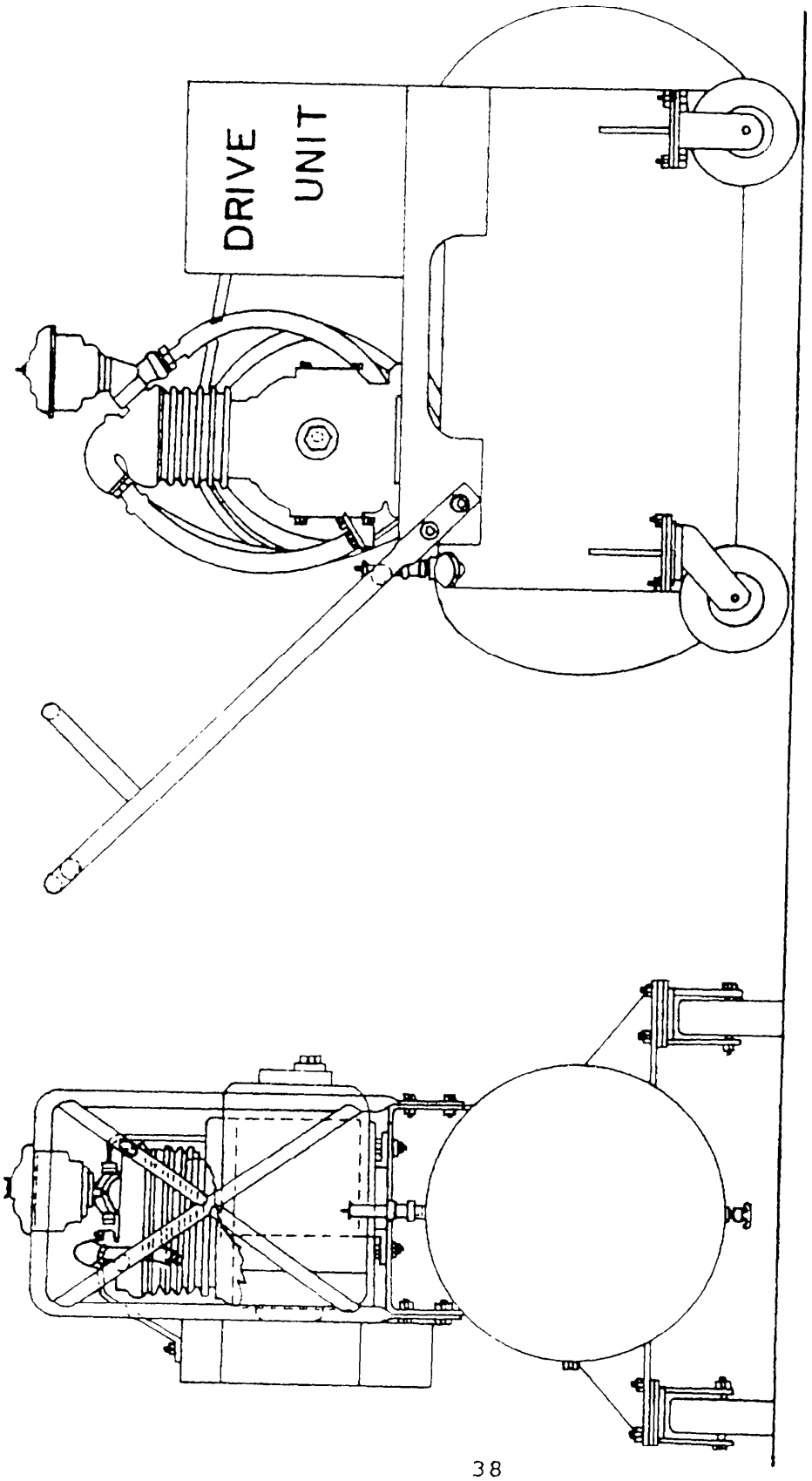


FIGURE 2. Compressor, air, mounted (general configuration).

X3641

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DATA SHEET
MFG NAME_____ P/N_____ Record Time & Amount of 1. Engine Oil Added 2. Compressor Oil Added Sheet ____ of____

Test Title:_____ Run No. _____ Date _____

Item	SYM	UNITS	Readings 1	2	3	4	5	6	7	8	9	10	...	...	
1. Time Start, Readings, Stop	R_t	HRS. MIN.	(IF STOP IS UNSCHEDULED GIVE REASON FOR STOP AND DOWN TIME)												
2. Temp. (Ambient) Dry Bulb	T_a	°F													
3. Temp. (Ambient) Wet Bulb	T_{wb}	°F													
4. Pressure (Ambient)	P_b	°F													
5. Temp., Engine Coolant	T_{e1}	°F													
6. Temp., Engine Lubricant	T_{e2}	°F													
7. Temp., Compressor Lubricant	T_L	°F													
8. Temp., Receiver Discharge	T_r	°F													
9. Pressure, Rec.	T_r	°F													
10. Pressure, Compressor Intake	Ps	°F													
11. Temp., Compressor Intake	T_1	°F													
12. Temp., Upstream of Nozzle	T_2	°F													
13. Pressure Differential Across Nozzle	dP	In.H_2O													
14. Compressor Speed	N	RPM													

FIGURE 3. First article test log.

X-19048

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-0052980C(ME)	2. DOCUMENT DATE (YYMMDD) 940811

3. DOCUMENT TITLE
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4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) ARMY BELVOIR RDE CTR W SATBE TSE 9101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340	DSN 289-2340

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